Applying the Basics

14. Fabricating Systems A production plant has two fabricating systems, both of which are maintained at 2-week intervals. However, one system is twice as old as the other. The number of finished products fabricated daily by each of the systems is recorded for 30 working days, with the results given in the table. Do these data present sufficient evidence to conclude that the variability in daily production warrants increased maintenance of the older fabricating system? Use the p-value approach.

v_2 New System	v_1 Old System
$\bar{x}_1 = 246$	$\bar{x}_2 = 240$
$s_1 = 15.6$	$s_2 = 28.2$

Q14 ① $H_0: \sigma_2^2 \le \sigma_1^2$

$H_1: \sigma_2^2 > \sigma_1^2$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{s_2^2}{s_1^2} \sim F_{29,29}$

④ $F^* = \frac{28.2^2}{15.6^2}$

$\simeq 3.268$

⑤ P-value based on p-val $F_{29,29} \sim F_{30,30} = 0.0006$

0.006 < 0.05

⑥ Since the p-value < α, we reject H_0.

16. SAT Scores The SAT subject tests in chemistry and physics for two groups of 15 students each electing to take these tests are given here.

v_1 Chemistry	v_2 Physics
$\bar{x} = 784$	$\bar{x} = 758$
$s = 114$	$s = 103$
$n = 15$	$n = 15$

To use the two-sample t-test with a pooled estimate of σ^2, you must assume that the two population variances are equal. Test this assumption using the F-test for equality of variances. What is the approximate p-value for the test?

Q16 $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$

$s_p^2 = \frac{14 \times 114^2 + 14 \times 103^2}{28}$

$s_p^2 = 11882.5$

① $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \ne \sigma_2^2$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{s_1^2}{s_2^2} \sim F_{14,14}$

④ $F^* = \frac{114^2}{103^2} = 1.225$

⑤ P-val approach $F_{14,14} \sim F_{5,14} = 0.69$

⑥ Since the p-val of $F_{14,14} > 0.05$, we do not reject H_0.

This make pooled estimate valid for t-test

22. Stock Risks The closing prices of two common stocks were recorded for a period of 15 days. The means and variances are

v_2	v_1
$\bar{x}_1 = 40.33$	$\bar{x}_2 = 42.54$
$s_1^2 = 1.54$	$s_2^2 = 2.96$

a. Do these data present sufficient evidence to indicate a difference between the variabilities of the closing prices of the two stocks for the populations associated with the two samples? Give the p-value for the test and interpret its value.

b. Construct a 99% confidence interval for the ratio of the two population variances.

Q22 a) ① $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \ne \sigma_2^2$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{s_1^2}{s_2^2} \sim F_{14,14}$

④ $F^* = \frac{2.96}{1.54} = 1.922$

⑤ P-val using P-val approach

$F_{14,14} = F_{15,14} = 0.1165$

⑥ Since the p-val of $2 \times P(F_{14,14}) > 0.05$, we fail to reject H_0.

b) 99% CI.

$\left[\frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{n_1-1, n_2-1; \alpha/2}}, \frac{s_1^2}{s_2^2} F_{n_2-1, n_1-1; \alpha/2}\right] = \left[1.922 \cdot \frac{1}{4.35}, 1.922 \cdot 4.35\right]$

$= [0.452, 8.1685]$

Since 1 lies within the range, we can conclude that the variances are equal.

DATA SET DS1040

11. Reaction Times A comparison of reaction times (in seconds) for two different stimuli in an experiment produced the following results when applied to a random sample of 16 people:

	x_1				x_2			
Stimulus 1	1	3	2	1	2	1	3	2
Stimulus 2	4	2	3	3	1	2	3	3

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

independent

$\bar{x}_1 = \frac{\sum x_i}{n_1}$

$= 1.875$

$s_1^2 = \frac{\sum_i (x_i - \bar{x}_1)^2}{n_1 - 1}$

$= 0.696$

$\bar{x}_2 = \frac{\sum_i x_i}{n_2}$

$= 2.625$

$s_2^2 = \frac{\sum_i (x_i - \bar{x}_2)^2}{n_2 - 1}$

$= 0.839$

① $H_0: \bar{x}_1 = \bar{x}_2$

$H_1: \bar{x}_1 \neq \bar{x}_2$

② $\alpha = 0.05$

③ $T_{stat} = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

④ $T^* = \frac{2.625 - 1.875}{0.839\sqrt{\frac{1}{8} + \frac{1}{8}}}$

$= 1.712$

$\frac{0.839}{0.696} = 1.205 < 3$

use pooled estimate

$s_p^2 = \frac{(8-1)0.839 + (8-1)(0.696)}{14}$

$s_p^2 = 0.7675$

$s_p = 0.876$

⑤ R.R.

$P_{14;0.025} = 2.145$

⑥ Since $T^* < P_{14;0.025}$, we do not reject H_0.

DATA SET DS1041

12. Reaction Times II Refer to Exercise 11. Suppose that the experiment is conducted using eight people as blocks and making a comparison of reaction times within each person; that is, each person is subjected to both stimuli in a random order. The reaction times (in seconds) for the experiment are as follows:

Person	Stimulus 1	Stimulus 2	
1	3	4	-1
2	1	2	-1
3	1	3	-2
4	2	1	1
5	1	2	-1
6	2	3	-1
7	3	3	0
8	2	3	-1

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

Q12 paired

$n = 8$

$\bar{d} = (-1-1-2+1-1-1+0-1)/8$

$= -0.75$

$s_d = 0.886$

① $H_0: \bar{d} = 0$

$H_1: \bar{d} \neq 0$

② $\alpha = 0.05$

③ $t_{stat} = \frac{\bar{d}}{s_p \sqrt{n}} \sim t_{(n-1)}$

④ $t^* = \frac{-0.75}{0.886\sqrt{8}}$

$= -2.40$

⑤ R.R.

$t_{7;0.025/2} = \pm 2.365$

⑥ Since $|t^*| > t_{7;0.025}$, we reject H_0